

# SENATE AI WORKSHOP

## Strategic LLM applications for institutional transformation

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# The Starting Formula

TASK + CONTEXT + OUTPUT FORMAT

**Task:** What you need done

**Context:** Information the LLM needs to do it well

**Output format:** How you need the results delivered



- "Focus only on programs with enrollment below 50 students."
- "Do not hedge. If the data says a program is failing, say so."

# Beyond the Formula

TASK + CONTEXT + OUTPUT FORMAT + **CONSTRAINTS** + ROLE

"Do not suggest strategies requiring budget increases"

"Focus only on programs with enrollment below 50 students"

"Do not hedge. If the data says a program is failing, say so."





- "You are a skeptical accreditation reviewer."
- "Focus only on programs with enrollment below 50 students."
- "You are a hiring manager screening accounting graduates."
- "Do not hedge. If the data says a program is failing, say so."
- "You are a grant reviewer looking for reasons to reject this proposal."

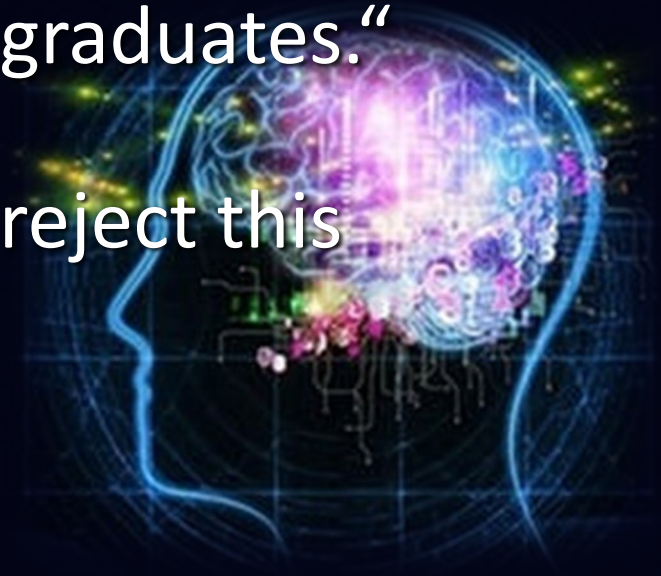
# Beyond the Formula

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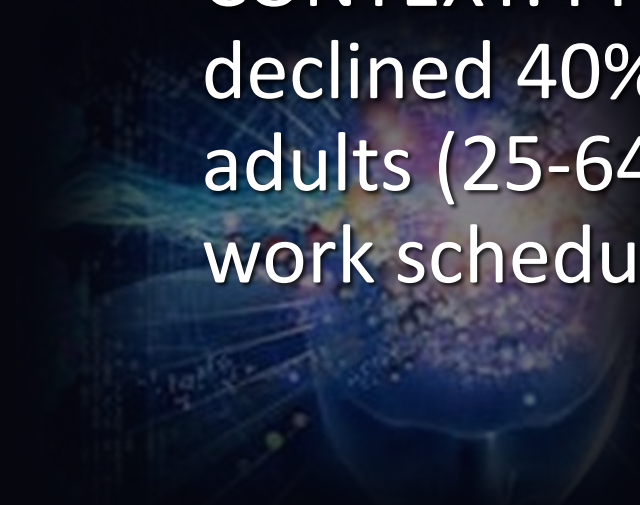
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# Context Is Intelligence

TASK: Analyze retention strategies for our campus.

CONTEXT: Private university in Puerto Rico. Enrollment declined 40% from 2020 to 2024. 49% of students are working adults (25-64). 70% are Pell-eligible. Top withdrawal reason is work schedule conflicts.

An abstract graphic in the bottom-left corner consisting of a blue and white particle trail or nebula-like shape, with small white dots and lines radiating from a central point.

# Three Signs of a Bad Prompt

**Generic output** — you forgot to include context about your institution

**Unusable output** — you forgot to specify constraints (budget, timeline, resources)

**Consultant-speak** — you asked for "best practices" instead of solving a specific problem



# The Enrollment Question

| Year | Total Enrollment | Change |
|------|------------------|--------|
| 2020 | 6,826            |        |
| 2025 | 4,101            | -40%   |

Source: IPEDS public data



# The Landscape

**Foundation Models and providers** GPT-4o, GPT-4.5, o1, o3, o4-mini (OpenAI). Claude Opus, Sonnet, Haiku (Anthropic). Gemini Ultra, Pro, Flash (Google). Llama 3, 4 (Meta). Mistral Large, Medium, Small, Codestral (Mistral AI). Command R+ (Cohere). Grok (xAI). DeepSeek-V3, DeepSeek-R1. Qwen 2.5, QwQ (Alibaba). Phi-4 (Microsoft)

**AI Coding and Development** GitHub Copilot. Claude Code. Cursor. Windsurf. Aider. Amazon Q Developer. Cline. Replit Agent. Devin. v0 (Vercel). Bolt. Lovable.

**Image, Video, and Audio Generation** GitHub Copilot. Claude Code. Cursor. Windsurf. Aider. Amazon Q Developer. Cline. Replit Agent. Devin. v0 (Vercel). Bolt. Lovable.





# The Landscape

**Retrieval, Agents, and Orchestration** RAG (Retrieval-Augmented Generation). LangChain. LlamaIndex. CrewAI. AutoGen. Model Context Protocol (MCP). Function calling. Tool use. Agentic workflows. Multi-agent systems. Computer use. Memory systems.

**Fine-Tuning and Customization** LoRA. QLoRA. RLHF. DPO. Constitutional AI. System prompts. Custom GPTs. Claude Projects. Gems (Gemini). Prompt caching. Context windows (128K, 200K, 1M+ tokens).

**Infrastructure and Deployment** Hugging Face. Ollama. vLLM. GGUF quantization. AWS Bedrock. Azure OpenAI. Google Vertex AI. Groq. Together AI. Fireworks AI. Modal. Replicate.

# The Landscape

**Evaluation and Safety** Red teaming. Prompt injection. Jailbreaking. Hallucination detection. Benchmarks (MMLU, HumanEval, GPQA, ARC-AGI). Alignment. Interpretability. Watermarking. Content authenticity.

**Emerging Capabilities (2025-2026)** Reasoning models (chain-of-thought, extended thinking). Computer use agents. Deep research agents. Real-time voice mode. Multimodal understanding (text + image + audio + video). Code execution sandboxes. Structured outputs. Citations and grounding.

This list is already outdated



- **Recognition** that the gap between what you know and what exists is widening despite the effort.
- **Dread** that falling behind carries professional consequences.
- **Exhaustion** from the effort coupled with the sense of futility.

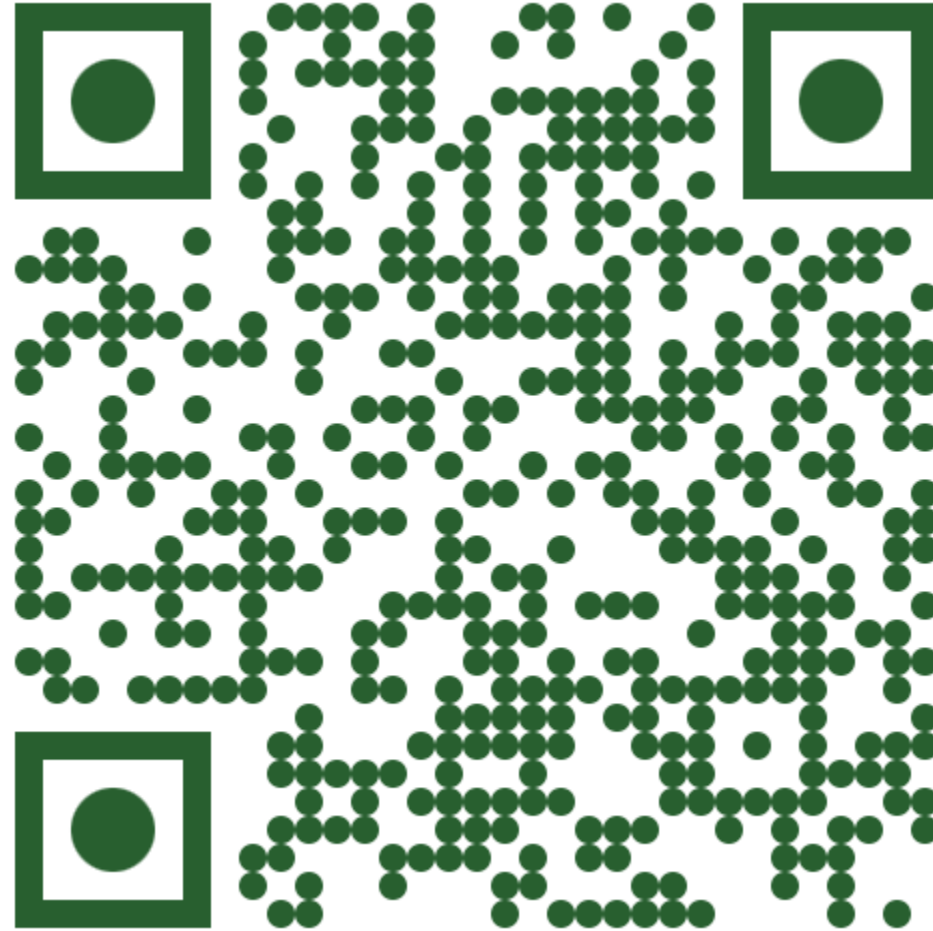
# AI Burnout

1. **Awareness** that something important is changing
2. **Effort** to keep up — reading, experimenting, attending workshops
3. **Recognition** that the gap between what you know and what exists is widening despite the effort
4. **Dread** that falling behind carries professional consequences
5. **Exhaustion** from the effort coupled with the sense of futility

- Recognition that the gap between what you know and what exists is widening despite the effort.
- Dread that falling behind carries professional consequences.
- Exhaustion from the effort coupled with the sense of futility.

## AI Burnout

1. Better thinking produces better AI output
2. Better AI output confirms the value of investing more thought
3. The high return on thinking makes it addictive
4. Hours disappear. 30 minutes becomes 4 hours. A workday in continuous strategic-thinking-plus-AI-execution is more cognitively demanding than most people expect.
5. The exhaustion is real but difficult to name



<https://github.com/pwayland/The-Repository>  
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